



AgriBean: A Simple bean Farming Guidance Application for Farmers

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1 Introduction

Bean (*Phaseolus vulgaris*) is one of the most important vegetable crops in Bangladesh, valued not only for its nutritional richness but also for its significant contribution to the rural economy. Despite its potential, many farmers face challenges in accessing structured, reliable, and timely information on Bean cultivation. Traditional farming practices often lack scientific guidance, and farmers are left without adequate support in areas such as disease management, pest control, and market access.

To bridge this knowledge gap, we propose **AgriBean**, a mobile application designed specifically for Bean farmers. The application is developed using the Flutter framework and incorporates an SQLite database to ensure offline functionality. AgriBean is envisioned as a farmer-friendly digital companion that delivers step-by-step cultivation guidance, practical solutions to common farming problems, and essential market-related insights. By combining simplicity with accessibility, the app aims to empower farmers with the knowledge and tools needed to improve productivity, reduce losses, and enhance commercialization opportunities.

2 Bean Cultivation in Bangladesh - An Overview

2.1 History

Bean (*Phaseolus vulgaris*) has a long history in Bangladesh and is considered the vegetable of the country. Historical records suggest that Bean has been cultivated in the Bengal region for centuries, both as a staple food and as a source of timber. Traditionally, Bean trees were planted in homestead gardens, village boundaries, and alongside roads, making them an integral part of rural life. The vegetable has been valued not only for its nutritional richness but also for its cultural significance, often being served during festivals, family gatherings, and community feasts. Over time, Bean cultivation has shifted from subsistence-level household production to more structured farming practices, with increasing attention from agricultural researchers and policymakers.

2.2 Production and Contribution

Bangladesh is one of the leading producers of Bean in the world. According to agricultural reports, the country produces over a million metric tons of Bean annually, making it one of the most abundant seasonal vegetables. Bean contributes significantly to the rural economy in several ways:

- **Nutritional Value:** Bean is rich in carbohydrates, vitamins (A, B, C), minerals, and dietary fiber, serving as a vital food source for rural households.

- **Economic Contribution:** Farmers earn substantial seasonal income from selling Bean in local and regional markets. The vegetable is also exported in processed forms such as chips, canned pulp, and pickles.
- **Employment:** Cultivation, harvesting, transportation, and processing of Bean generate employment opportunities for rural communities.
- **Timber and By-products:** Bean wood is widely used in furniture-making and construction, while leaves and other by-products are used as fodder for livestock.

Despite its importance, post-harvest losses remain high due to inadequate storage, transportation, and processing facilities. Addressing these challenges could further enhance Bean’s contribution to food security and the national economy.

2.3 Major Bean-Growing Regions

Bean is cultivated across Bangladesh, but certain regions are particularly well-known for their production due to favorable soil and climatic conditions. The major Bean-growing regions include:

- **Gazipur and Narsingdi:** Known for large-scale Bean orchards and commercial production.
- **Tangail and Mymensingh:** Significant contributors to both household consumption and market supply.
- **Chattogram Hill Tracts:** Produces high-quality Bean due to fertile soils and favorable rainfall.
- **Jessore and Khulna:** Important for both Bean and timber production.
- **Sylhet Region:** Bean is grown alongside tea and other crops, contributing to the region’s mixed agricultural economy.

These regions not only supply local markets but also transport Bean to urban centers such as Dhaka and Chattogram, ensuring wide availability during the harvesting season. With proper storage, processing, and value-chain development, these regions have the potential to become hubs for Bean-based agro-industries.

2.4 Challenges of Bean Cultivation in Bangladesh

Although Bean is the national vegetable of Bangladesh and holds immense economic and nutritional value, its cultivation faces several challenges that limit productivity, quality, and commercialization. These challenges can be categorized into agricultural, environmental, economic, and infrastructural aspects:

- **Agricultural Challenges:**

- *Lack of Scientific Cultivation:* Most Bean trees are grown in homestead areas without systematic orchard management, leading to inconsistent yields.
- *Pest and Disease Infestation:* Common problems include stem borers, vegetable rot, and fungal infections, which reduce both quality and quantity of produce.
- *Poor Post-Harvest Handling:* Farmers often lack knowledge of proper harvesting techniques, resulting in vegetable damage and reduced shelf life.

- **Environmental Challenges:**

- *Climate Vulnerability:* Bean cultivation is highly sensitive to irregular rainfall, drought, and storms, which are becoming more frequent due to climate change.
- *Soil Degradation:* Declining soil fertility in some regions reduces productivity and increases the need for improved soil management practices.

- **Economic Challenges:**

- *Market Instability:* Prices fluctuate widely during the peak season, often forcing farmers to sell at low rates due to oversupply.
- *Limited Value Addition:* Lack of processing facilities means most Bean is sold fresh, leading to high post-harvest losses and missed opportunities for value-added products such as chips, canned pulp, or pickles.
- *Middlemen Dependency:* Farmers often rely on intermediaries for selling their produce, reducing their profit margins.

- **Infrastructural and Institutional Challenges:**

- *Poor Storage and Transportation:* Inadequate cold storage and inefficient transport systems result in significant post-harvest losses.
- *Limited Research and Extension Services:* Farmers have limited access to updated scientific knowledge, training, and expert guidance on modern Bean cultivation techniques.
- *Policy Gaps:* Despite being the national vegetable, Bean receives less institutional support compared to other cash crops, limiting investment in research, marketing, and export promotion.

Summary

In summary, while Bean cultivation has great potential in Bangladesh, it is constrained by a combination of agricultural inefficiencies, environmental risks, economic barriers,

and infrastructural shortcomings. Addressing these challenges through improved farming practices, better post-harvest management, market development, and stronger institutional support is essential for unlocking the full potential of Bean as a driver of rural development and food security.

2.5 Consumption and Export Potential of Bean in Bangladesh

Bean is not only the national vegetable of Bangladesh but also one of the most widely consumed seasonal vegetables in the country. Its versatility, nutritional richness, and cultural significance make it an essential part of rural and urban diets. At the same time, Bean holds considerable potential for export, both in fresh and processed forms, which could contribute to the national economy if properly harnessed.

Domestic Consumption

- **Nutritional Role:** Bean is a rich source of carbohydrates, vitamins (A, B, C), minerals, and dietary fiber. It is consumed as a staple vegetable during the summer season and provides an affordable source of nutrition for rural households.
- **Culinary Uses:** The vegetable is consumed in multiple forms:
 - *Ripe Bean:* Eaten fresh as a sweet vegetable or used in desserts.
 - *Unripe Bean:* Cooked as a vegetable curry, often referred to as the “poor man’s meat” due to its texture and nutritional value.
 - *Seeds:* Boiled, roasted, or ground into flour for various dishes.
- **Cultural Significance:** Bean is often served during family gatherings, festivals, and community feasts, making it an integral part of Bangladeshi food culture.
- **By-products:** Leaves are used as livestock fodder, while Bean wood is valued for furniture and construction, further extending its domestic utility.

Export Potential

- **Global Demand:** There is a growing international demand for Bean, particularly in countries with large South Asian and Southeast Asian communities. In recent years, Bean has also gained popularity in Western markets as a plant-based meat substitute due to its fibrous texture.
- **Processed Products:** Beyond fresh vegetable, Bean can be exported in processed forms such as:
 - Frozen Bean bulbs and pulp.
 - Canned Bean in syrup or brine.

- Bean chips, pickles, and flour.
- Ready-to-cook or vacuum-packed unripe Bean for vegan markets.
- **Challenges in Export:** Despite its potential, Bangladesh faces several barriers:
 - Lack of modern processing and packaging facilities.
 - High post-harvest losses due to poor storage and transportation.
 - Limited international branding and marketing of Bangladeshi Bean.
- **Opportunities:** With investment in cold storage, processing plants, and quality certification, Bangladesh could position Bean as a competitive export commodity. Partnerships with international food companies and promotion of Bean-based vegan products could open new markets in Europe, North America, and the Middle East.

Summary

In summary, Bean plays a dual role in Bangladesh: it is a vital source of nutrition and cultural identity domestically, while also holding untapped potential as an export-oriented crop. By addressing post-harvest challenges and investing in value-added processing, Bangladesh can transform Bean from a seasonal vegetable into a globally recognized agricultural product.

3 Objectives and Scope

3.1 Objectives

The AgriBean application has been developed with the following key objectives:

- To provide structured, step-by-step guidance on Bean cultivation, covering land preparation, planting, irrigation, fertilization, pest control, and harvesting.
- To offer practical solutions for common farming challenges, including pest infestations, plant diseases, and environmental stress factors.
- To ensure the application remains simple, intuitive, and usable by farmers with little or no technical background, while functioning effectively in offline environments.
- To connect farmers with essential market-related information, enabling them to make informed decisions about pricing, selling, and commercialization of Bean products.

3.2 Scope

The scope of the AgriBean application extends to the following areas:

- Providing comprehensive guidance on cultivation techniques, seasonal care, and harvesting practices tailored to the local agricultural context of Bangladesh.
- Storing all essential information in an SQLite database to ensure offline accessibility, particularly in rural areas with limited or no internet connectivity.
- Delivering a simple and user-friendly interface that prioritizes clarity, readability, and ease of navigation for non-technical users.
- Laying the foundation for future enhancements, including integration with real-time weather updates, online market platforms, and potentially AI-driven advisory features to further support decision-making.

4 Stakeholders' Requirement

4.1 Stakeholders

The AgriBean mobile application is designed to serve a diverse range of stakeholders, each with unique requirements and expectations. By addressing these needs, the app ensures maximum usability, impact, and long-term sustainability.

- **Farmers:** Require simple, step-by-step farming guidance, problem-solving features for pests and diseases, offline access to cultivation data, and timely market insights to make informed selling decisions.
- **Agricultural Experts:** Need a reliable platform to share validated, research-based information with farmers, ensuring that best practices and scientific knowledge are effectively disseminated.
- **Students/Researchers:** Seek structured agricultural data, case studies, and real-world insights into Bean cultivation for academic projects, research papers, and innovation in agri-tech.
- **Entrepreneurs:** Require information about production volumes, supply chain opportunities, and commercialization potential to identify business prospects in Bean farming and processing.
- **Government/NGOs:** Need a communication channel to promote sustainable farming practices, monitor adoption of agricultural policies, and support rural development initiatives.

- **Traders/Market Agents:** Require access to production and availability data to connect with farmers, streamline procurement, and stabilize supply-demand chains.
- **Extension Workers:** Need a digital tool to assist in training and advising farmers in rural areas, ensuring consistent delivery of agricultural knowledge even in remote regions.
- **Technology Developers:** Require a scalable and modular system architecture that allows future integration of advanced features such as weather forecasting, AI-driven crop diagnosis, and online marketplaces.
- **Policy Makers:** Need aggregated data and usage analytics from the app to understand farming trends, identify challenges, and design evidence-based agricultural policies.
- **Local Communities:** Benefit indirectly through improved agricultural productivity, better income opportunities, and enhanced food security resulting from the app's adoption.

4.2 RACI Matrix

The following RACI Matrix outlines the roles and responsibilities of key stakeholders in the AgriBean project:

Stakeholder	Responsible	Accountable	Consulted	Informed
Farmers	Yes	No	No	Yes
Agricultural Experts	No	Yes	Yes	No
Developers (AgriBean Team)	Yes	Yes	No	Yes
Government / NGOs	No	Yes	Yes	No
Traders / Market Agents	No	No	Yes	Yes
Students / Researchers	No	No	Yes	Yes

Table 2: RACI Matrix for AgriBean Project in Bangladesh

5 Requirement Gathering

Requirement gathering is a critical step in the development of the AgriBean mobile application. It ensures that the system is designed to meet the needs of its primary users (farmers) while also being technically feasible, sustainable, and adaptable for future enhancements. The requirements are divided into two categories: functional and non-functional.

5.1 Functional Requirements

The functional requirements define the core features and services that the AgriBean application must provide to its users:

- **Cultivation Guidance:** Farmers must be able to view step-by-step instructions on Bean cultivation, including land preparation, planting, irrigation, fertilization, pest management, and harvesting tips.
- **Problem–Solution Module:** The app should include a dedicated section where farmers can browse common farming challenges (e.g., pest infestations, diseases, soil issues) and receive practical, validated solutions.
- **Offline Data Storage:** An SQLite database must be integrated to store all essential information locally, ensuring that farmers in rural areas with limited or no internet connectivity can still access the app’s content.
- **User-Friendly Navigation:** The interface should be simple and intuitive, with clear icons, minimal text, and support for the local language (Bangla) to ensure accessibility for non-technical users.
- **Market Information:** The app should provide farmers with basic market insights, such as price trends and commercialization opportunities, to help them make informed selling decisions.
- **Feedback Mechanism:** Farmers should be able to provide feedback on the app’s content and usability, enabling continuous improvement and adaptation to user needs.

5.2 Non-Functional Requirements

The non-functional requirements define the quality attributes and constraints that ensure the AgriBean application is efficient, reliable, and sustainable:

- **Usability:** The app must prioritize simplicity, with a clean layout, clear icons, and minimal text. Instructions should be presented in farmer-friendly language, ensuring accessibility for users with low literacy or limited technical skills.
- **Performance:** The system should be optimized to load quickly and operate smoothly, even on low-end smartphones with limited processing power and memory. Database queries and navigation between sections should be efficient.
- **Reliability:** Offline access is essential to ensure uninterrupted usability. The app must function without internet connectivity, with data stored locally and synchronized when a connection becomes available.

- **Maintainability:** The system should be designed with modular architecture, making it easy for developers to update or extend features, such as adding new farming guidelines, pest solutions, or market modules.
- **Scalability:** Although initially focused on Bean farming, the app should be designed in a way that allows future expansion to other crops or integration with additional services (e.g., weather updates, online marketplaces).
- **Security:** Basic security measures must be in place to protect user data, including login credentials and stored information, ensuring that farmers' personal data is not misused.
- **Localization:** The app should support Bangla as the primary language, with the possibility of adding other regional languages in the future to broaden accessibility.

6 System Development Process

There are many ways to develop a system. For the development of the **AgriBean Mobile Application**, we followed a structured approach based on the **Software Development Life Cycle (SDLC)**. The following steps were undertaken to ensure systematic progress and quality assurance.

6.1 System Life Cycle

The development of the **AgriBean Mobile Application** follows the **Software Development Life Cycle (SDLC)**, ensuring a structured and systematic approach. The key phases include:

- **Planning:** In this phase, the scope, objectives, and feasibility of the AgriBean application were defined. Stakeholders such as farmers, agricultural experts, and NGOs were identified, and their needs were analyzed. A project plan was created, including risk assessment, resource allocation, and a timeline for deliverables.
- **Requirement Analysis:** Both functional and non-functional requirements were gathered. Functional requirements included cultivation guidance, pest and disease solutions, and offline data storage. Non-functional requirements emphasized usability, performance on low-end devices, and offline reliability.
- **System Design:** The system architecture was designed using Flutter for cross-platform development and SQLite for offline storage. The design phase also included database schema creation, user interface mockups, and defining navigation flows to ensure simplicity for non-technical farmers.

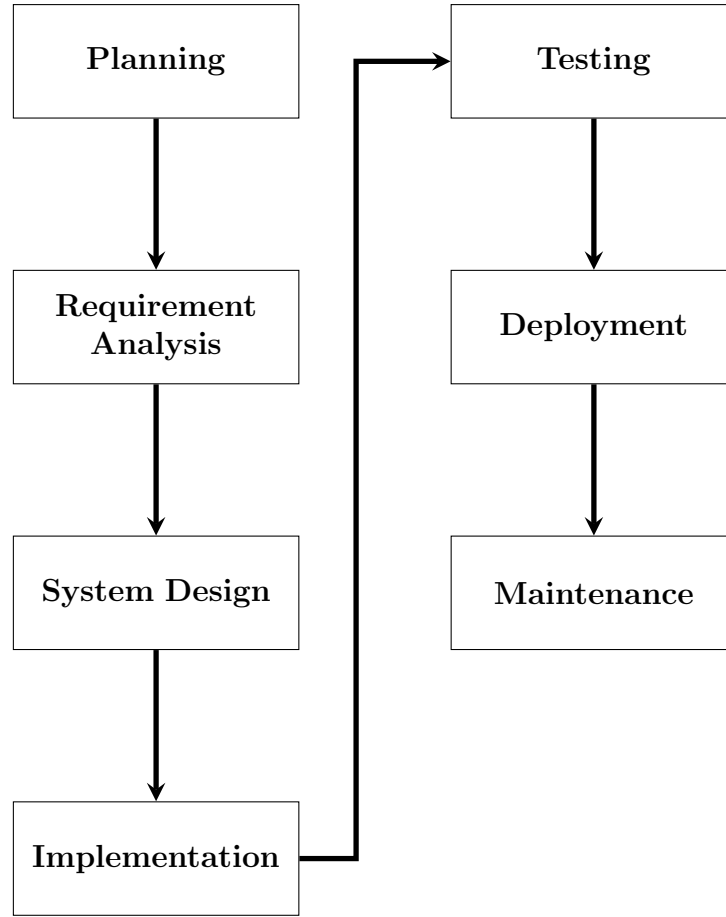


Figure 1: System Development Process for AgriBean

- **Implementation:** The mobile application was developed using the Flutter framework with Dart programming language. SQLite was integrated for local data storage, and the interface was designed with farmer-friendly icons, Bangla language support, and minimal text for ease of use.
- **Testing:** Multiple levels of testing were conducted, including unit testing of individual modules, integration testing of the database and UI, and user acceptance testing with sample farmers to ensure the app met real-world needs.
- **Deployment:** The application was deployed on Android devices, as this platform is most widely used by farmers in Bangladesh. The deployment process included packaging the app for distribution and ensuring compatibility with low-end smartphones.
- **Maintenance:** After deployment, the app is monitored for performance, bug fixes, and user feedback. Future updates will include new farming guidelines, integration with weather APIs, and potential online market features.

The adoption of SDLC ensures that AgriBean is developed systematically, reduces risks, and guarantees that the final product is reliable, user-friendly, and aligned with the

needs of Bangladeshi farmers.

7 Application Architecture

7.1 Entity–Relationship Model

The Entity–Relationship (ER) Model of the **AgriBean Mobile Application** defines the logical structure of the database and illustrates how different entities interact with each other. The model ensures that all essential data related to farmers, cultivation guidance, problem–solution mapping, and market information is systematically stored and retrieved.

7.1.1 Entities and Attributes

- **User (Farmer):** Represents the primary users of the system.
 - Attributes: *UserID (PK), Name, Age, Location, ContactInfo, LanguagePreference*
- **Guidance:** Contains structured information on Bean cultivation.
 - Attributes: *GuidanceID (PK), StepTitle, Description, Season, Category (Cultivation / Pest Control / Harvesting)*
- **ProblemSolution:** Stores common farming problems and their recommended solutions.
 - Attributes: *ProblemID (PK), ProblemName, Symptoms, Solution, SeverityLevel*
- **MarketInfo:** Provides commercialization and market-related insights.
 - Attributes: *MarketID (PK), ProductName, PriceRange, DemandLevel, MarketTips*
- **Feedback:** Captures user feedback for continuous improvement.
 - Attributes: *FeedbackID (PK), UserID (FK), Comments, Rating, Date*

7.1.2 Relationships

- A **User** can access multiple **Guidance** records. (1:N)
- A **User** can report multiple **Problems**, each linked to a **ProblemSolution**. (M:N, resolved via junction table)
- A **User** can view multiple **MarketInfo** entries. (1:N)

- A **User** can submit multiple **Feedback** entries. (1:N)
- Each **ProblemSolution** may be linked to one or more **Guidance** steps for detailed instructions. (M:N)

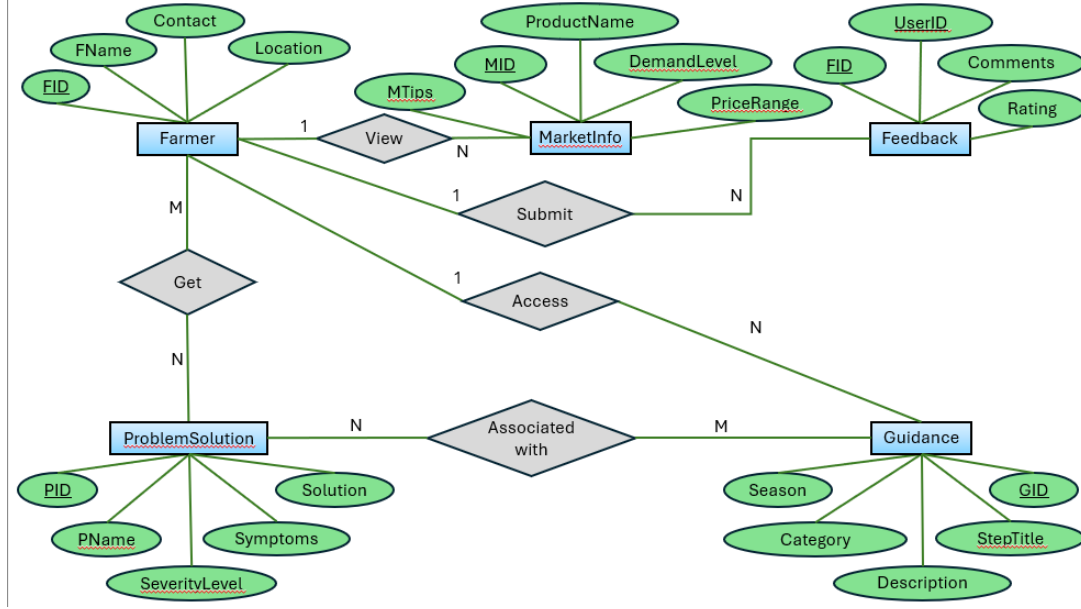


Figure 2: Entity–Relationship Diagram of AgriBean

7.1.3 Description

The ER Model ensures that:

- Farmers' details are securely stored and linked to their activities within the app.
- Cultivation guidance is categorized and easily retrievable.
- Problems reported by farmers are mapped to validated solutions provided by agricultural experts.
- Market information is accessible to help farmers make informed selling decisions.
- Feedback from farmers is collected for iterative improvements of the application.

7.2 3-Layer Model

The **AgriBean Mobile Application** is designed using a three-layer architecture, which separates the system into distinct layers to improve maintainability, scalability, and clarity of responsibilities. Each layer has a specific role in ensuring smooth operation and user experience:

1. **Presentation Layer:** This layer is implemented using the Flutter framework and provides the user interface through which farmers interact with the application. It includes screens such as the home dashboard, problem reporting forms, solution display pages, and the marketplace view. The design emphasizes simplicity, local language support, and intuitive navigation so that even non-technical users can easily access farming guidance and market information.
2. **Business Logic Layer:** This layer acts as the core of the application, handling workflows and decision-making processes. It manages the mapping between reported problems and their corresponding solutions, processes user inputs, and ensures that the correct guidance or market data is delivered to the farmer. By isolating the logic from the interface, this layer ensures that the app remains flexible and easier to update when new features or rules are introduced.
3. **Data Layer:** The data layer is responsible for storing and retrieving information. It uses an SQLite database to maintain offline access to cultivation steps, problem-solution records, and market insights. This ensures that farmers in rural areas with limited or no internet connectivity can still benefit from the application. The data layer also supports synchronization with online services in future versions, allowing updates such as real-time weather data or live market prices.

This layered approach ensures that the application is modular, easy to maintain, and capable of evolving with future enhancements while keeping the user experience simple and reliable.

7.3 Data Flow Diagram Model

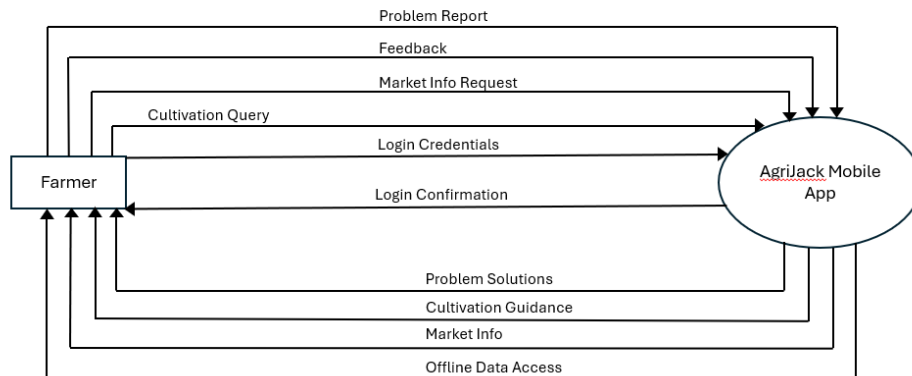


Figure 3: Data Flow Diagram (Level 1) of AgriBean

8 Application Workflow

8.1 Functionality Flowchart

Figure 4: Functionality Flow of AgriBean

8.2 User Interface Workflow

The user interface (UI) of the **AgriBean Mobile Application** is designed with simplicity and accessibility in mind, ensuring that farmers with limited technical knowledge can easily navigate the system. The workflow follows a logical sequence of screens and interactions, guiding the user from the home screen to specific features such as problem reporting, solution retrieval, and market insights. The major steps are as follows:

1. Home Screen:

- Serves as the entry point of the application after login or offline access.
- Displays quick access tiles for major categories such as *Cultivation Guidance*, *Problem Reporting*, and *Marketplace*.
- Provides rotating highlights of seasonal tips, best practices, and urgent alerts (e.g., pest outbreaks).
- Includes a simple navigation bar with icons for *Home*, *Guidance*, *Problems*, *Market*, and *Feedback*.

2. Problem Reporting:

- Farmers can select their problem from a categorized list (e.g., pests, diseases, soil issues).
- Each problem entry includes a short description and common symptoms to help farmers identify the issue.
- Farmers may optionally provide additional notes or feedback for better personalization.
- The app stores reported problems locally in SQLite for offline access and later synchronization.

3. Solutions:

- Once a problem is selected, the app retrieves the corresponding solution from the *ProblemSolution* database.

- Solutions are displayed in a step-by-step format, with simple text and icons for clarity.
- Where applicable, the app links to related *Guidance* entries for more detailed cultivation practices.
- Farmers can mark solutions as *helpful* or *not helpful*, feeding into the feedback mechanism for continuous improvement.

4. Marketplace:

- Provides farmers with basic market insights such as current price ranges, demand levels, and commercialization tips.
- Displays information in a simple tabular or card format for easy comparison.
- Offers offline access to previously stored market data, ensuring usability even without internet connectivity.
- Future versions may integrate real-time updates from online market platforms and weather APIs.

Workflow Summary

The workflow ensures that farmers can:

- Start from a simple and intuitive home screen.
- Report problems quickly and receive immediate, validated solutions.
- Access cultivation guidance and seasonal tips without technical barriers.
- Stay informed about market opportunities to improve commercialization.

This design prioritizes **usability, offline reliability, and farmer empowerment**, making AgriBean a practical digital companion for Bean farmers in Bangladesh.

8.3 Source Code

The app is developed using Flutter framework. SQLite is integrated for offline database operations.

- `main.dart`: Entry point of the application.
- `db_helper.dart`: Handles SQLite operations.
- `ui/`: Contains user interface screens for different modules.

9 Conclusion and Future Work

The **AgriBean Mobile Application** has been conceptualized and developed as a practical, farmer-centric solution to address the challenges faced by Bean growers in Bangladesh. By combining structured cultivation guidance, problem–solution mapping, and offline accessibility through SQLite, the system ensures that even farmers in remote areas with limited internet connectivity can benefit from reliable agricultural knowledge. The simple and intuitive interface, designed with local language support, makes the application accessible to non-technical users, thereby bridging the gap between modern agricultural practices and traditional farming communities.

The development of AgriBean demonstrates how mobile technology can empower farmers by providing them with timely information, validated solutions, and market insights. Beyond improving productivity and reducing crop losses, the system also contributes to the broader goals of food security, rural development, and sustainable agriculture. The integration of feedback mechanisms ensures that the application can evolve continuously, adapting to the real needs of its users.

Future Work

While the current version of AgriBean provides a strong foundation, several enhancements can be incorporated to expand its functionality and impact:

- **Real-time Weather Integration:** Incorporating weather APIs will allow farmers to receive timely updates on rainfall, temperature, and storm warnings, enabling them to make informed decisions about irrigation, fertilization, and harvesting.
- **Online Marketplace:** Future versions can include a digital marketplace that connects farmers directly with buyers, traders, and cooperatives. This feature would reduce dependency on middlemen, ensure fair pricing, and open new commercialization opportunities for Bean products.
- **Multilingual Support:** Expanding the application to support multiple regional languages will make it more inclusive and accessible to farmers across different parts of Bangladesh and potentially other Bean-growing regions.
- **AI-driven Advisory System:** Machine learning models can be integrated to provide personalized recommendations based on farmer location, soil type, and historical problem reports, making the app more adaptive and intelligent.
- **Integration with Government and NGO Platforms:** Linking AgriBean with agricultural extension services, subsidy programs, and NGO initiatives will strengthen its role as a trusted channel for policy implementation and farmer support.

- **Scalability to Other Crops:** Although initially designed for Bean, the system architecture can be extended to cover other crops, thereby broadening its impact and ensuring long-term sustainability.

Summary

In conclusion, AgriBean is not just a mobile application but a step toward digital transformation in agriculture. By addressing the immediate needs of Bean farmers and laying the groundwork for future enhancements, it has the potential to become a comprehensive agricultural support system that empowers farmers, strengthens rural economies, and promotes sustainable farming practices in Bangladesh and beyond.